1.3 Transformations of Linear and Absolute Value Functions

Name _____ Period ____ Date _____

Algebra II 1.3 · Student sheet

I can…

- write functions that represent translations and reflections.
- write functions that represent stretches and shrinks.
- write functions that represent combinations of transformations.

Vocabulary · fill in the missing word

Horizontal translation — g(x) = _____

Vertical translation — g(x) = _____

Reflection in the x-axis / y-axis — g(x) = _____ = f(−x)

Vertical stretch or shrink — g(x) = _____

Horizontal stretch or shrink — g(x) = _____ when a > 1, a stretch when 0 < a < 1

Part 1 · Predict, then check

Predict the vertex and describe the change from f(x) = |x|. Then check in Desmos.

Function	Predicted vertex	Description	Correct? (✓ or fix)
|x − 2|			
|x| − 2			
|2x|			
2|x|			
−|x| + 1			

Part 2 · Notes and examples (fill in each note, then work the example)

Inside or outside: Horizontal changes go inside: f(_____). Vertical changes go outside: _____.

Translations and reflections: Right h: f(_____) Up k: _____ Reflect in the y-axis: f(_____)

Example 1

f(x) = 2x − 1. Write a rule for g, a translation 3 units up, and for h, a translation 2 units left.

Example 2

f(x) = |x + 3| + 1. Write a rule for g, a reflection in the x-axis, and for h, a reflection in the y-axis.

Horizontal shrink: Horizontal shrink by 1/a: f(_____)

Example 3

a. $f(x) = x + 2$. Write g, a horizontal shrink by a factor of $\frac{1}{3}$.

b. $f(x) = |x| - 1$. Write h, a vertical stretch by a factor of 4.

Example 4

$f(x) = x$. Write g, a vertical stretch by a factor of 2 followed by a translation 1 unit left and 4 units down.

Part 3 · Partner practice

Partner 1

$f(x) = 3x + 4$. Write g, a translation 5 units right.

Partner 2

$f(x) = |x|$. Write g, a reflection in the x-axis followed by a translation 3 units up.